

Grade 6
Unit 7
Lesson 15 Practice

Name Answer Key
Date _____
Period _____

Evaluate each expression.

1. $(3)^5 = 243$

$$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$$

2. $(-3)^3 = -27$

$$(-3)(-3)(-3)$$

3. $\left(-\frac{2}{3}\right)^4 = \frac{16}{81}$

$$\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)$$

4. $\left(\frac{1}{3}\right)^5 = \frac{1}{243}$

$$\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)$$

5. The expression $(-6)^3$ will result in a ☐ negative ☐ positive value because

the ☐ base ☐ exponent is an ☐ odd ☐ even number.

For questions 6-10, select all equivalent values or expressions.

6. $(1.8)^4 = (1.8)(1.8)(1.8)(1.8)$

- ☐ 7.2
- ☐ $(4)(1.8)$
- ☐ 10.4976
- ☐ $(3.24)(3.24)$
- ☐ $1.8 \cdot 1.8 \cdot 1.8 \cdot 1.8$

7. $(-3)^5$

- ☐ -15
- ☐ -243
- ☐ $(5)(-3)$
- ☐ $(9)(-3)$
- ☐ $(-3)(-3)(-3)$

8. $\left(\frac{1}{4}\right)^3$ $\left(\frac{1}{4}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{64}$

- ☐ $\frac{1}{12}$
- ☐ $\frac{1}{64}$
- ☐ $\left(\frac{1}{16}\right)(3)$
- ☐ $\left(\frac{1}{16}\right)\left(\frac{1}{4}\right)$
- ☐ $\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4}$

9. $(-5)^4$

- ☐ 625
- ☐ $(25)^2$
- ☐ -625
- ☐ $(-25)(-25)$
- ☐ $(-5)(-5)(-5)(-5)$

10. $(-9)(-9)(-9) = -729$

- ☐ $(9)^3$
- ☐ 729
- ☐ $(-9)^3$
- ☐ -729
- ☐ $-9 \cdot (-9)^2$